

# Input-Affine Conditioning of Selective State-Space Models Is Gauge-Absorbable: A Diagnostic and a Design Rule

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Paper under double-blind review

## Abstract

Selective state-space models such as Mamba have reshaped long-context sequence modeling. Yet which forms of conditioning *survive* their training dynamics is poorly understood. We study FiLM-style regime conditioning of a Mamba backbone: a per-block, channel-wise affine on each block’s input. This is the natural drop-in way to steer the input-dependent selective-scan parameters  $\Delta, \mathbf{B}, \mathbf{C}$ . We find a **decisive null** with an actionable mechanistic explanation. Under joint training on weakly decodable market regimes, the modulation collapses near identity ( $\text{film}_g \approx 0.004\text{--}0.010$ ) across FI-2010 equities, Kaggle crypto spot books, and Polymarket prediction markets, and it cannot be forced active even by direct supervision. The scale component of an input-side affine is a **gauge** direction: it folds losslessly into the block’s own  $\Delta/\mathbf{B}/\mathbf{C}$  projections. In the Mamba3 blocks used here, the additive shift has no matching bias that can absorb it in full (only its  $\Delta$ -component can fold, into the  $dt$  bias) and decays as an unrewarded component. Forced-active FI-2010/Kaggle runs decay monotonically with descriptive order-of-thousands exponential fits ( $R^2 > 0.999$ ). An output-side placement shows the same split: scale is absorbed into the bias-free output projection, while the shift is unrewarded and also decays. The usable consequence is negative and diagnostic. Do not expect input- or output-affine regime conditioning of a selective scan to persist unless the regime is strongly decodable. Before crediting any conditioning, verify the gate is active. An inert gate reported beside a positive generalization gap means the gap is an *artifact*, not a conditioning effect. The positive design rule is scoped: a surviving modulation should act on  $\Delta/\mathbf{B}/\mathbf{C}$  *inside* the fused kernel, a target validated at toy reference-scan scale (seeds 0–3) but not yet demonstrated at full world-model or foundation-model scale.

## 1 Introduction

Conditioning a network on side information (a task ID, a context vector, an inferred regime) is among the most common things a practitioner does, and FiLM (a channel-wise affine  $\gamma \odot x + \beta$ ) (Perez et al., 2018) is the default mechanism. When the backbone is a **selective state-space model** (Mamba (Gu & Dao, 2024)), there is an especially tempting place to inject it. Mamba’s scan parameters  $\Delta_t, \mathbf{B}_t, \mathbf{C}_t$  are *input-dependent* linear projections of each block’s input. An affine on that input therefore flows straight into the dynamics, and, crucially, it needs **no change to the fused CUDA/TileLang kernel**. This is the drop-in form everyone reaches for first. We show it does not work, explain mechanistically why, and turn that explanation into two things a practitioner can use: a diagnostic that catches a specific false-positive, and a design rule that says where a surviving modulation has to live.

**The finding.** We introduce a *RegimeFiLM* modulator that infers a categorical latent regime and emits per-block affine parameters  $(\gamma, \beta)$ . It is zero-initialized, so at step 0 the modulated backbone is identical to the unmodulated baseline (an exact paired control). We then apply the strongest activation pressure we could: forced-active initialization plus direct

supervision of the router. The modulation still decays to identity on every market objective and dataset we test. The cause is a **gauge symmetry** for the multiplicative component. A constant channel-wise scale on the block input folds losslessly into the host projections  $W_\Delta/W_B/W_C$  as a right-multiplied diagonal  $W_\bullet \text{diag}(\gamma)$ . Mamba3’s input projections are bias-free up to a lone  $dt$  bias that can absorb only the shift’s  $\Delta$ -component, so the additive shift is not gauge in our implementation; it has no rewarded role in these weakly decodable regimes and decays as an unrewarded component. Joint training therefore returns  $(\gamma, \beta) \rightarrow (1, 0)$  by a scale-gauge plus unrewarded-shift mechanism. Forced-active FI-2010 and Kaggle runs decay monotonically, with order-of-thousands descriptive time constants ( $R^2 > 0.999$ ) that are similar across datasets because the scale-absorption direction is a property of the *block parameterization*, not the data. Moving the gate to the block *output* does not escape this: the scale folds into the bias-free output projection, and the residual shift again earns nothing.

**Why this is actionable.** Two upshots change what a practitioner does.

- *A diagnostic that catches a false-positive.* The activation diagnostics  $\text{film}_g = \text{mean}|\gamma - 1|$  and the router entropy  $\text{reg}_H$  (uniform maximum log  $R$ ) are **load-bearing**. In our data a generalization-gap metric reads *positive*, with the “right” sign, while FiLM is provably inert and the treatment is *absolutely worse* in both regimes. The gap is a distributional (high – low) artifact whose sign even flips with head calibration. A practitioner who reports “conditioning improved the out-of-regime gap” without checking that the gate is actually *on* can ship an artifact.
- *A design rule that says where to put the gate.* If you want regime conditioning of a selective scan to *survive* training, the immediately actionable negative rule is to skip input- and output-affine placements when the router is uniform. The constructive placement must act **between the block’s bounding projections**, modulating  $\Delta/B/C$  inside the fused kernel. We validate that target at toy reference-scan scale; we do not claim a full world-model or foundation-model in-kernel implementation here.

**Demonstration substrate.** To stress the mechanism on genuinely different data we build a Mamba-3 world model for limit-order-book (LOB) microstructure in the DreamerV3/Drama lineage (Hafner et al., 2023; Wang et al., 2024; Lahoti et al., 2026), exercised on FI-2010 equities, Kaggle crypto spot books, and Polymarket prediction markets (§4). The backbone is a substrate, not a claim. Under matched parameters, *no* backbone dominates across datasets (§5.3), exactly what a result about how selective scans *train*, rather than which one is best, should predict. The same study yields a deflated economic coda (§5.3) that a logistic regression on the same features reproduces, so it is not attributable to the architecture.

## 2 Related Work

**Selective state-space models and Mamba.** Structured state-space models (S4, S5, Mamba) give  $O(n)$ -complexity alternatives to Transformers (Gu et al., 2022; Smith et al., 2023; Gu & Dao, 2024). Mamba introduced *selective* state spaces in which  $\Delta$  and  $\mathbf{B}$  are input-dependent, enabling content-aware filtering; Mamba-2 (Dao & Gu, 2024) formalized the duality with structured masked attention, and Mamba-3 (Lahoti et al., 2026) adds multi-input/multi-output (MIMO) transitions. The property we exploit is that  $\Delta_t, \mathbf{B}_t, \mathbf{C}_t$  are learned linear projections of the block input, and this is exactly what makes the scale component of an input-affine a gauge direction. We are not aware of prior work characterizing the training-time fate of FiLM-style conditioning of a selective scan.

**FiLM and feature-wise conditioning.** FiLM (Perez et al., 2018) injects context as a channel-wise affine and is the default conditioning primitive across vision, multi-task, and regime-embedding settings (Lim et al., 2021). Our result is about *where* in a selective-scan block an affine can be placed and still survive joint training. This is a gauge constraint specific to the input-dependent projection structure, orthogonal to FiLM’s expressive-power story.

**Interpretability of training dynamics and weight symmetries.** A line of interpretability work studies *how* solutions form rather than *what* a fixed network computes, including reparameterization and gauge symmetries that leave the function unchanged while reshaping the loss landscape (Dinh et al., 2017). In the normalization literature, a channel-wise affine before a linear map is known to be redundant with that map. van Laarhoven (2017) shows weights feeding a normalization layer are scale-invariant, so  $L_2$  regularization acts on their *scale*, and thus on the effective learning rate, rather than as capacity control; Hoffer et al. (2018) and Li & Arora (2020) re-derive weight decay as an effective-learning-rate schedule on the same symmetry. Our null is the same redundancy in a new setting. The absorbed component is not a static normalization gain but the scale of a per-regime, *input-dependent* FiLM that folds into the selective scan’s own  $W_\Delta/W_B/W_C$  projections. What is new is (i) the input-dependent selective-scan instantiation; (ii) the observed monotone forced-active decay, with similar order-of-thousands descriptive fits ( $R^2 > 0.999$ ; architecture-general by the algebra of the fold, not yet shown at FM scale), attributing the collapse to the optimizer; (iii) the output-side placement showing the same scale-gauge plus unrewarded-shift split; and (iv) the practical consequence that the only kernel-friendly, drop-in form is exactly the absorbable one. The redundancy is therefore the default a practitioner reaches for, not a corner case. The actionable contribution is to turn that symmetry into a measurable diagnostic ( $\text{film}_g, \text{reg}_H$ ) and a placement rule.

**LOB models and regime-switching (substrate).** DeepLOB (Zhang et al., 2019) and successors are discriminative LOB models on FI-2010 (Ntakaris et al., 2018); generative LOB work (Nagy et al., 2023; Backhouse et al., 2025) targets simulation fidelity; classical regime-switching and Bayesian changepoint detection identify discrete states from price processes (Hamilton, 1989; Adams & MacKay, 2007). We use these only as the demonstration substrate and benchmark context, not as baselines for the conditioning result.

### 3 Setup: FiLM Conditioning of a Selective Scan

**Selective SSM.** A discretized state-space layer maps an input  $x_t \in \mathbb{R}^d$  to an output  $y_t$  via a latent state  $h_t$ :  $h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t$ ,  $y_t = \mathbf{C}h_t$ , where  $\bar{\mathbf{A}}, \bar{\mathbf{B}}$  are the discretized state-transition and input matrices and  $\mathbf{C}$  the output matrix. Mamba makes the parameters **input-dependent**:

$$\Delta_t = \sigma(W_\Delta x_t), \quad \mathbf{B}_t = W_B x_t, \quad \mathbf{C}_t = W_C x_t, \quad (1)$$

where  $W_\Delta, W_B, W_C$  are learned projection matrices,  $\sigma$  is the softplus nonlinearity,  $\Delta_t$  is the per-step discretization size, and  $\mathbf{B}_t, \mathbf{C}_t$  are the now input-dependent scan parameters, so *any affine transformation of  $x_t$  propagates directly into all three scan parameters*. Mamba-3 further makes the state transition  $A_t$  a data-dependent projection of  $x_t$ , so the affine propagates there too. This is the selection mechanism we attempt to steer.

**RegimeFiLM modulator.** A regime head produces a soft categorical  $\alpha_t = \text{softmax}(W_r x_t^{(0)}) \in \mathbb{R}^R$  over  $R$  regimes ( $R = 4$  for FI-2010/Kaggle and  $R = 8$  for Poly-market), with router weights  $W_r$  acting on the stem output  $x_t^{(0)}$ . It reads out a regime embedding  $r_t = \alpha_t^\top E$ , where  $E$  is a learned regime-embedding table; a hypernetwork maps  $r_t$  to per-block affine parameters, bounded for bf16 stability:

$$\gamma^{(\ell)} = 1 + \tanh(\tilde{\gamma}^{(\ell)}) \in (0, 2), \quad \beta^{(\ell)} = \tanh(\tilde{\beta}^{(\ell)}) \in (-1, 1), \quad (2)$$

where  $\ell$  indexes the block,  $\gamma^{(\ell)}, \beta^{(\ell)}$  are that block’s FiLM scale and shift, and  $\tilde{\gamma}^{(\ell)}, \tilde{\beta}^{(\ell)}$  are the hypernetwork’s raw (pre-tanh) outputs. The hypernetwork is **zero-initialized**, so at step 0  $\gamma^{(\ell)} = \mathbf{1}, \beta^{(\ell)} = \mathbf{0}$  for all  $\ell$  and the modulated backbone is identical to the unmodulated baseline, a clean and exact paired control. A Switch-Transformer load-balancing term (Fedus et al., 2022) maximizes the entropy of the *batch-averaged* regime distribution,  $\mathcal{L}_{\text{regime}} = \lambda_{\text{ent}}(\log R - H[\bar{\alpha}])$  (with  $\lambda_{\text{ent}}$  a weight,  $H[\cdot]$  the entropy, and  $\bar{\alpha}$  the batch-averaged router distribution), regularizing marginal usage while leaving per-step assignments free to be peaky.

140 **Two application sites.** The **input-affine** site routes the regime into the scan through the  
 141 selection mechanism:

$$x_t^{(\ell)} = x_t^{(\ell-1)} + \text{Mamba3}(\gamma^{(\ell)} \odot \text{RMSNorm}(x_t^{(\ell-1)}) + \beta^{(\ell)}), \quad (3)$$

142 where  $x_t^{(\ell)}$  is block  $\ell$ 's output (input  $x_t^{(\ell-1)}$ , the previous block's output),  $\odot$  the element-wise  
 143 product,  $\text{RMSNorm}$  RMS normalization, and  $\text{Mamba3}(\cdot)$  the selective-scan block of Eq. (1).  
 144 As a control we also expose a **post-scan** site that gates the block *output*:

$$x_t^{(\ell)} = x_t^{(\ell-1)} + \gamma^{(\ell)} \odot \text{Mamba3}(\text{RMSNorm}(x_t^{(\ell-1)})) + \beta^{(\ell)}. \quad (4)$$

145 Both share the identical modulator; only the site differs. This isolates the gauge structure  
 146 (§5): the input-side scale folds into  $W_\Delta/W_B/W_C$ , the post-scan scale folds into the bias-free  
 147 output projection  $W_{\text{out}}$ , and the additive shift cannot be absorbed in full in either placement.

148 **Diagnostics.** Activation metrics gate every claim:  $\text{film}_g = \text{mean}|\gamma - 1|$  (deviation of the  
 149 multiplicative gate from identity), its additive counterpart  $\text{film}_b = \text{mean}|\beta|$  (reported for  
 150 the output-side shift in §5.2), and the router entropy  $\text{reg}_H$  (uniform maximum  $\log R$ , with  $R$   
 151 the regime count;  $\log 4$  on FI-2010/Kaggle,  $\log 8$  on Polymarket). **A positive generalization**  
 152 **gap with an inert FiLM is an artifact, not a regime-conditioning effect.** This is the central  
 153 methodological point.

## 154 4 Demonstration Substrate (condensed)

155 **Limit order books, for a non-finance reader.** A *limit order book* (LOB) is the live ledger  
 156 an exchange keeps for an asset. At each instant it records how much volume rests on the  
 157 buy and sell sides at each price near the mid-price, and it updates many times a second  
 158 as orders arrive, cancel, and execute. The task is to forecast the next short-horizon move  
 159 (price direction, volatility, or order flow) from the changing shape of this ladder. We span  
 160 three *structurally different* kinds of book on purpose, so that one shared null across them  
 161 speaks to the optimizer rather than to a single dataset. **FI-2010** is regulated equities (Nasdaq  
 162 Nordic): centralized, fixed-hours, and discrete-tick, with mature and comparatively stable  
 163 microstructure. **Crypto** spot books trade 24/7 with no circuit breakers across many venues,  
 164 giving heavier tails and thinner, faster-moving books; we sample them at 1 s/1 min/5 min  
 165 resolutions. **Polymarket** is a binary prediction market rather than a continuous price. Each  
 166 contract settles to 0 or 1 at a known expiry, so its price is a probability in  $[0, 1]$  set by a  
 167 time-to-expiry clock and the reference move since open. Equity microstructure, crypto  
 168 volatility, and a bounded settling probability are deliberately far apart.

169 To exercise the mechanism on genuinely different distributions we instantiate the modulated  
 170 backbone inside a Mamba-3 world model for LOB microstructure (DreamerV3/Drama  
 171 lineage (Hafner et al., 2023; Wang et al., 2024)): a Transformer-over-depth encoder feeds a  
 172  $16 \times 16$  categorical posterior and an  $L=4$  stack of Mamba-3 SISO blocks ( $d_{\text{state}}=128$ ), trained  
 173 with the standard DreamerV3 loss (reconstruction + dynamics/representation KL) plus  
 174 light auxiliary heads. The rest of the machinery (episodic memory, a Hawkes head, the  
 175 imagination loop, the EV-head internals) is confined to an anonymized supplementary  
 176 implementation; no conditioning result depends on any of it.

177 **Regime axis.** Beyond realized volatility we use a **predictability** axis on the underlying  
 178 spot series (the asset's live market price). Its primary measure is the Kaufman efficiency  
 179 ratio  $\text{ER} = |s_n - s_0| / \sum_t |s_t - s_{t-1}|$ , where  $s_t$  is the spot price at step  $t$  within the window  
 180 and  $s_0, s_n$  are its first and last values; ER is near 1 for a clean trend and near 0 for a random  
 181 wander. We also use permutation entropy and the Hurst exponent. These measures serve  
 182 two roles. They split the held-out evaluation into forecastable versus random-walk regimes,  
 183 and, when the router collapses, they let us *supervise it directly*. "Is this window forecastable"  
 184 is the most defensible regime signal we have. A router that will not encode even this axis is  
 185 the strongest form of the null.

| Family            | Objective       | Dataset    | Gap (sign, unit)        | film <sub>g</sub> | Verdict |
|-------------------|-----------------|------------|-------------------------|-------------------|---------|
| Volatility/scale  | StudentT NLL    | FI-2010    | $\approx 0$ nats        | $\approx$ ident.  | NULL    |
| Order-flow volume | Volume NLL      | FI-2010    | $\approx 0$ nats        | $\approx$ ident.  | NULL    |
| Direction         | macro-F1        | FI-2010    | +0.008 <sup>†</sup> F1  | 0.018             | NULL    |
| Settlement        | BCE log-loss    | Polymarket | +0.18 <sup>†</sup> nats | 0.008             | NULL    |
| <b>PnL</b>        | simulated trade | Polymarket | −\$627 <sup>†</sup>     | 0.010             | NULL    |

Table 1: RegimeFiLM is inert across every objective.  $\text{reg}_H$  equals its uniform maximum ( $\log R$ ) in all rows. The “Gap” column is in each objective’s own units (reconstruction NLL nats, macro-F1 points, BCE log-loss nats, dollars), so its magnitude is not comparable across rows; only its sign and the inert  $\text{film}_g$  are. <sup>†</sup>a non-zero gap whose sign is a distributional artifact, not a FiLM effect (FiLM is inert, and the treatment is absolutely worse). The PnL (profit-and-loss) row’s −\$627 is the FiLM A/B gap on the PnL *objective* (treatment−control), *not* an economic total (cf. the +\$5,147 settlement total in §5.3). The direction-row  $\text{film}_g = 0.018$  is the endpoint of the *supervised* escalation on that head (the joint-trained value is  $\approx 0.004$ , cf. Table 2); the settlement (0.008) and PnL (0.010) rows are joint-trained. All are inert, but the column mixes the two measurement types.

**Datasets.** (i) Polymarket binary-outcome crypto LOBs (BTC/ETH/SOL up/down at 5/15/60-min intervals, with the underlying Chainlink/Binance spot path); (ii) the public FI-2010 benchmark (Ntakaris et al., 2018); (iii) Kaggle high-frequency crypto spot books (BTC/ETH/ADA at 1 s / 1 min / 5 min). Held-out markets are split by predictability (primary) and realized volatility (secondary), both reported. **All thresholds are fit on the training split and frozen before any held-out evaluation**, and baseline and treatment share the seed (an exact paired control).

## 5 The Null and Its Mechanism

### 5.1 RegimeFiLM is a decisive null

Table 1 reports the central diagnostic across five objectives spanning two datasets and a money objective. In every family FiLM collapses to identity under joint training:  $\text{film}_g$  decays toward zero and  $\text{reg}_H$  sits at its uniform maximum ( $\log R$ ). Where a generalization gap reads positive (direction +0.008, settlement +0.18, by sign), FiLM is provably inert, the treatment is *absolutely worse* in both regimes, and the gap is a high − low distributional artifact whose sign even flips with head calibration, which is exactly why the activation diagnostics are load-bearing. Read *without* the gate diagnostic, both gaps carry the sign a successful conditioner would produce, and we would have reported a spurious win. Only the inert  $\text{film}_g$  exposes the gap as an artifact, and this is the reading we ourselves nearly published.

**Polymarket forced-active supervision test.** Forcing FiLM active on Polymarket (zero-init scale raised so  $\text{film}_g = 0.150$  at step 0) and supervising the router *directly* on the spot efficiency-ratio axis with the strongest settings (LR multiplier 50, supervision weight 30, decoupled router/FiLM, fed observation volatility) still decays the modulation monotonically toward identity ( $\text{film}_g$ : 0.150  $\rightarrow$  0.119  $\rightarrow$  0.100  $\rightarrow$  0.074  $\rightarrow$  0.056  $\rightarrow$  0.043, still falling), with  $\text{reg}_H = \log 8$  throughout. The capacity is present but unused. On FI-2010, a router head trained alone on the *volatility* bucket reaches 0.92 agreement, a capacity check (a linear probe on the frozen representation is weaker, App. B), yet even there the modulation collapses. On the *predictability* axis, where this escalation supervises the router, the regime decodes only weakly ( $\approx 0.25$ –0.36).

### 5.2 The mechanism: scale gauge plus unrewarded shift

**Cross-dataset collapse points at the optimizer.** We take the architecture finding to FI-2010 (re-examined under the predictability-primary axis) and the Kaggle crypto spot book



| Dataset    | Asset / resolution | Gap (pred / vol)        | $\text{film}_g$ | Verdict |
|------------|--------------------|-------------------------|-----------------|---------|
| FI-2010    | event stream       | $\approx 0 / \approx 0$ | 0.004           | NULL    |
| Kaggle BTC | 1 min              | $\approx 0 / \approx 0$ | 0.004           | NULL    |
| Kaggle ETH | 1 min              | $\approx 0$             | 0.005           | NULL    |
| Kaggle ADA | 1 min              | $\approx 0$             | 0.005           | NULL    |
| Kaggle BTC | 5 min              | $\approx 0$             | 0.005           | NULL    |
| Kaggle BTC | 1 s (slice)        | $\approx 0$             | 0.005           | NULL    |

Table 2: Cross-dataset confirmation of the regime-FiLM null.  $\text{reg}_H = \log 4$  (uniform router) in every row. The null is dataset-, asset-, and resolution-independent.

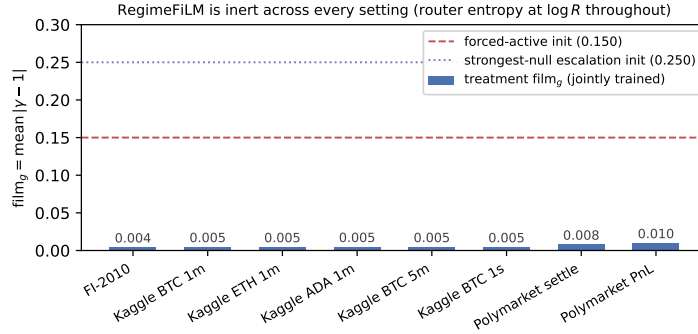


Figure 1: **RegimeFiLM is inert across every A/B setting**, the gap-positive-but-gate-off diagnostic this paper is built on. The jointly-trained treatment  $\text{film}_g = \text{mean } |\gamma - 1|$  (bars) sits near 0.004–0.010 on every dataset, asset, and resolution (FI-2010, Kaggle BTC/ETH/ADA at three bar widths, and the two Polymarket objectives), while a forced-active initialization starts at 0.150 and the strongest-null escalation at 0.250 (dashed lines). The router entropy sits at its uniform maximum  $\log R$  throughout. The modulation never grows from its exact identity initialization.

(BTC/ETH/ADA at three resolutions), judged by forecasting metrics alone. The zero-initialized modulation never leaves identity:  $\text{film}_g \approx 0.004$  and  $\text{reg}_H = \log 4$  in every setting (Table 2). With FiLM an exact paired control, every held-out gap is statistically indistinguishable from zero (all 95% CIs include 0) on both regime axes. The split is non-degenerate. High-efficiency-ratio reference windows carry genuinely more structure on all three predictability measures, and DeepLOB attains 0.628 macro-F1 on the FI-2010 horizon-10 labels. So the null is a negative on *forecastable* data, not a no-signal artifact.

**The collapse is optimizer-driven.** The predictability-supervised strongest-null test (forced-active FiLM, router supervised directly on the efficiency-ratio bucket), applied at the **input-side** gate, decays  $\text{film}_g$  monotonically at every logged step on *both* datasets ( $0.250 \rightarrow 0.144$  Kaggle,  $0.250 \rightarrow 0.147$  FI-2010), with  $\text{reg}_H \approx \log 4$  throughout *even though the load-balance term is disabled*. The uniform router is therefore not a load-balance artifact. A descriptive exponential fit  $a e^{-t/\tau} + c$  matches both trajectories at  $R^2 > 0.999$  with similar order-of-thousands parameters ( $a \approx 0.30$ ,  $\tau \approx 6.7\text{--}7.2\text{k}$  steps). The decay is monotone, with a negative slope sustained to the last logged step (endpoint  $\text{film}_g \approx 0.15$ , still falling), and the fitted asymptote lies at or below identity ( $c \lesssim -0.05$ , only more negative when the lower bound on  $c$  is relaxed). We therefore rest the claim on the observed monotone decay and treat identity as the extrapolated limit, not as a precisely identified invariant. That the decay scale is similar across two very different markets points at the optimizer, not the data (Figure 2). Repeating this escalation on the *best-decoded* volatility bucket instead (the axis a frozen-router head fits to 0.92 on FI-2010) gives the same decay to 0.144 on both datasets ( $\tau \approx 6700/6610$ ,  $R^2 = 0.9999$ ), so the collapse is gauge absorption, not an undecodable supervised axis.

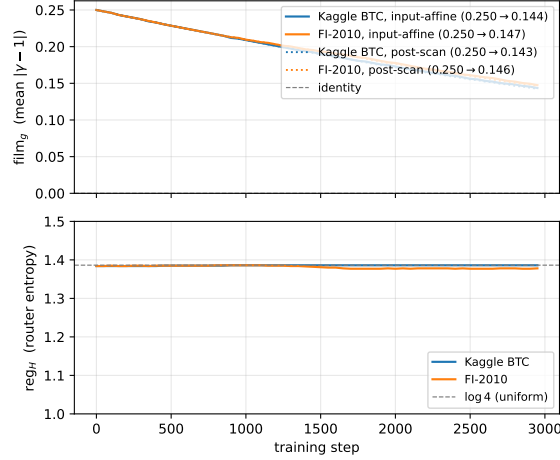


Figure 2: **Strongest-null test, and its placement comparison.** Forcing the modulation active ( $\text{film}_g = 0.250$  at step 0) and supervising the router directly on the efficiency-ratio bucket still decays  $\text{film}_g$  monotonically toward identity on both datasets (top, solid = *input-side* gate), with  $\text{reg}_H$  pinned at its uniform maximum  $\log 4$  (bottom). An exponential fit gives  $R^2 > 0.999$  with an order-of-thousands time constant and a negative fitted asymptote, so the modulation extrapolates toward identity rather than to a positive plateau. The *output-side* post-scan gate (top, dotted) decays along the same trajectory. Its scale folds into the bias-free output projection just as the input-side scale folds into  $W_\Delta / W_B / W_C$ ; in both placements the shift is unrewarded under the weakly decodable regime. Endpoints agree to plotting precision by placement (input-side  $0.250 \rightarrow 0.147$  FI-2010 /  $0.144$  Kaggle; output-side  $0.250 \rightarrow 0.146$  FI-2010 /  $0.143$  Kaggle, all at the last logged step).

**Proposition (absorbability criterion).** Let a channel-wise affine  $(\gamma, \beta)$  act on an activation immediately consumed by a learned linear map  $A$  with optional bias  $b$ . The *scale* is always gauge,  $A(\gamma \odot x) = (A \text{diag}(\gamma))x$ , absorbed into  $A$  with or without a bias. The *shift* is gauge only when  $A$  carries a trainable bias and no nonlinearity intervenes,  $A(x + \beta) + b = Ax + (b + A\beta)$ ; without such a bias it is non-absorbable. A mechanism that instead modifies the in-kernel scan parameters  $W_\Delta, W_B, W_C$ —or any quantity strictly between a block’s bounding projections—is *not* gauge-absorbable. *Corollary:* in the Mamba3 configuration used here (bias-free in/out projections; a lone  $dt$  bias that can absorb only the shift’s  $\Delta$ -component), input- and output-side FiLM share the same split: scale is gauge ( $W_{\{\Delta, B, C\}} \text{diag}(\gamma)$  or  $\text{diag}(\gamma)W_{\text{out}}$ ), while shift decays only as an unrewarded non-absorbable component.

**Constant vs. input-dependent components.** The criterion applies to the *constant* affine component induced when the router is uniform. A trained FiLM can also have an input-dependent component through the router. In weakly decodable regimes the supervised router collapses toward uniform, so the modulation reduces to a constant scale plus shift; the scale is gauge-absorbable and the shift is unrewarded. When the regime is strongly decodable, the input-dependent component can survive, as the LM control below illustrates.

**Why identity is a free direction: gauge absorption.** By the criterion, our input-side affine  $\hat{x} = \gamma \odot \hat{x} + \beta$  is consumed immediately by the re-projections  $\Delta_t = \sigma(W_\Delta \hat{x}_t)$ ,  $\mathbf{B}_t = W_B \hat{x}_t$ ,  $\mathbf{C}_t = W_C \hat{x}_t$ . Its scale folds into  $W_{\{\Delta, B, C\}}$  with **no change to the function the block computes**, so the host weights can represent any constant scale themselves. The shift would fold only in a biased projection; in the Mamba3 blocks used here (a lone  $dt$  bias can absorb only its  $\Delta$ -component), it is instead an additive component that the weakly decodable regime objective does not reward. Joint training therefore folds the scale into the host weights and decays the residual shift, returning  $(\gamma, \beta) \rightarrow (\mathbf{1}, \mathbf{0})$ , which is precisely the monotone

decay we observe. The similar order-of-thousands decay fits follow because the absorption direction is a property of the block parameterization, not the data. The actionable negative is that the most kernel-friendly, drop-in form of the idea is exactly the form the optimizer can and does undo when the router is uniform.

**An output-side placement: the same split.** The gauge argument names an obvious next test, gating *after* the scan. The output scale folds into the bias-free  $W_{\text{out}} (\gamma \odot (W_{\text{out}} h) = (\text{diag}(\gamma) W_{\text{out}}) h)$ , while the shift, with no output bias and a nonlinear RMSNorm before the next block, is genuinely **non-absorbable**. The experiment bears this out. Under the identical forced-active escalation, the output-side  $\text{film}_g$  and  $\text{film}_b$  decay in lockstep on both datasets ( $\text{film}_g$ :  $0.250 \rightarrow 0.146$  FI-2010,  $0.250 \rightarrow 0.143$  Kaggle) with the same order-of-thousands descriptive fits ( $\tau \approx 7065$  FI-2010 /  $6598$  Kaggle,  $R^2 > 0.999$ ), and input- and output-side endpoints differ only at plotting precision. The symmetry between placements is therefore scale gauge plus unrewarded shift: the host weights absorb the scale on either face, and the non-absorbable shift decays because the weakly decodable regime loss does not reward it. A modulation escaping both must act *between* the block’s bounding projections, modulating  $W_\Delta / W_B / W_C$  inside the fused kernel.

**The boundary, demonstrated (with a gauge-vs-regularization control).** Toy-scale experiments turn the design rule from asserted to demonstrated. (i) *Folding*. On a toy reference scan with biased projections, an input-affine modulation folds into the host projections to floating-point precision (fold residual  $1.2 \times 10^{-6}$ ) and decays to identity ( $\text{film}_g : 0.086 \rightarrow 0.010$ ). The *same* modulation applied to  $\Delta / B / C$  *inside* the scan is non-absorbable ( $1.3 \times 10^{-2}$ ,  $\sim 10^4 \times$  larger) and **persists** ( $0.093 \rightarrow 0.175$ , seeds 0–3), so the in-kernel target the rule names is real at reference-scan scale. In the Mamba3 runs (bias-free in/out projections), this fold validates the scale gauge; the additive shift is handled by the unrewarded-shift argument above. (ii) *Weight-decay control*. Removing weight decay flattens the decay ( $\tau$  from  $\approx 6685$  to  $\approx 3 \times 10^5$ ,  $R^2$   $0.9999 \rightarrow 0.48$ ), so the gauge-direction decay is driven by **weight decay over a flat loss**, not the objective. It is absent for the in-kernel placement in this toy setting; a full world-model or foundation-model in-kernel implementation remains future work.

**When conditioning *does* survive: regime decodability.** The collapse is not universal, and the boundary is itself actionable. A Mamba char-LM control on a two-domain corpus (English prose vs. Python source), with the regime inferred by a context-aware router supervised on the domain and the *same* input-affine FiLM, does **not** collapse. In this single-seed toy control, a strongly-decodable regime (router accuracy  $\geq 0.9$ ;  $0.91$  at the last logged step) makes  $\text{film}_g$  *rise* from  $0.11$  to  $0.52$ ; the qualitative boundary is the point, not the calibrated plateau value. Gauge absorption always removes the *constant scale* component. The *input-dependent* part survives only when the regime is decodable enough to be worth encoding. In the LOB settings the regime is weakly decodable, the router collapses to uniform, and the modulation reduces to a constant scale plus an unrewarded shift. **The actionable rule, refined:** skip the input-affine FiLM exactly when the regime is weakly decodable from the block input; if it is strongly and contextually decodable, the affine *can* engage.

### 5.3 Architecture-agnosticism check (matched-parameter backbone ablation)

We compare Mamba-1, Mamba-2, Mamba-3 (SISO), and a Transformer-over-time at matched budgets ( $\sim 10.3\text{M}$ ) on FI-2010 and the Kaggle BTC spot book, reporting held-out Student- $t$  reconstruction NLL (Table 3). The result is a genuine ambiguity: **no backbone leads on both datasets**. Mamba-1 wins FI-2010, Mamba-2 wins Kaggle, and Mamba-3 SISO is competitive but not strictly dominant. A pre-norm Transformer matched to the Mamba-3 block ( $9.61\text{M}$ ) is likewise competitive but non-dominant, so the non-dominance is not an under-tuned-baseline artifact. These single-seed runs show no clear dominant backbone in this setup; they are not a statistically resolved ranking.

*Why SISO.* We run Mamba-3 in SISO throughout. The MIMO kernel’s dynamic shared-memory footprint exceeds the available GPU cap at every warp-valid (rank, chunk) con-



| Backbone                | Params | FI-2010 recon NLL | Kaggle recon NLL |
|-------------------------|--------|-------------------|------------------|
| Mamba-1                 | 11.45M | −0.5403           | 0.0241           |
| Mamba-2                 | 10.20M | −0.5309           | −0.0213          |
| Mamba-3 SISO            | 10.30M | −0.5346           | 0.0207           |
| Transformer (post-norm) | 9.90M  | −0.5175           | 0.0191           |
| Transformer (pre-norm)  | 9.61M  | −0.5041           | 0.0224           |

Table 3: Matched-parameter backbone ablation (held-out reconstruction NLL, lower is better). Single-seed runs show no clear dominant backbone here, so the choice of sequence model is not the operative variable; this is not a statistically resolved ranking.

figuration, so the MIMO variant is outside this paper’s hardware envelope and we make no claim about it. This is the same kernel boundary the design rule names. The surviving, in-kernel modulation of  $\Delta/\mathbf{B}/\mathbf{C}$  lives behind the same fused-kernel wall.

**Deflated economic coda (substrate).** The substrate yields a positive result the architecture did *not* carry. The world model’s gated settlement probability is a survivable edge on the high-predictability settlement subset (market-block 95% CI [ $+\$2,445$ ,  $+\$7,747$ ] over 127 traded/evaluable markets), yet it is **architecture-independent**. On the full 645-market held-out BTC set, the settlement head totals  $+\$5,147$ ; a logistic on the *same* features exceeds it ( $+\$6,914$ , 243 traded/evaluable markets under the same gate). The operative variable is the features and the gate, **not the sequence model**; the 246-market GBT comparison is in Appendix A.

## 6 Discussion: What a Practitioner Should Do Differently

The interpretability finding is small: the scale component of an input-affine on a Mamba block input is a gauge direction. The lift-and-use protocol is:

### Diagnostic protocol: will my selective-scan conditioning survive training?

**1. Instrument the gate.** Log  $\text{film}_g = \text{mean}|\gamma - 1|$  and  $\text{reg}_H$ . Before crediting conditioning, require  $\text{film}_g > 0$  and  $\text{reg}_H < \log R$ ; a positive gap against an inert gate is a distributional artifact, not a conditioning effect.

**2. Run the fold-residual placement test.** On a tiny reference scan, fit the host projections to absorb the forced-active scale (and any shift only when the host projection has a trainable bias). Input- and output-affine scales fold to round-off ( $\sim 10^{-6}$ ); a placement survives only if its residual is large ( $\sim 10^{-2}$ ), i.e. it acts on  $\Delta/\mathbf{B}/\mathbf{C}$  between the block’s bounding projections.

**3. Check regime decodability.** Affine conditioning engages only when the regime is strongly decodable from the block input. If the router collapses to uniform, only a constant scale-plus-shift remains; the scale is gauge-absorbable and the shift is unrewarded, so skip the affine. (Established on a single-seed LM control, so treat the threshold as directional, not calibrated.)

**Limitations.** The forced-active decay is monotone over our training horizon and admits order-of-thousands descriptive fits with asymptotes at or below identity, but the within-horizon endpoint is still above identity; we do not rule out every positive plateau under every optimizer/schedule. The LM control is single-seed and the in-kernel toy is seeds 0–3, while the market null is multi-setting and multi-seed. The economic coda is a backtest substrate, not part of the interpretability claim.

## 7 Conclusion

An input-affine FiLM on a Mamba block input contains a **scale gauge**; its shift decays when the weakly decodable regime loss does not reward it. In our runs the affine silently returns to identity, and output-side placement shows the same split. The practitioner’s rule is the absorbability criterion: condition between a block’s bounding projections, on  $\Delta/\mathbf{B}/\mathbf{C}$  inside

the fused kernel (validated at toy scale, seeds 0–3), or gate every claim on the two-number  $\text{film}_g/\text{reg}_H$  check.

## Use of Large Language Models

In keeping with the COLM 2026 policy on disclosure of LLM use, we report that a large language model was used while preparing this paper. It assisted in drafting and editing the prose, including the phrasing of the reference list, and was used to assist with evaluation.

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## 384 A Deflated economic coda: figures

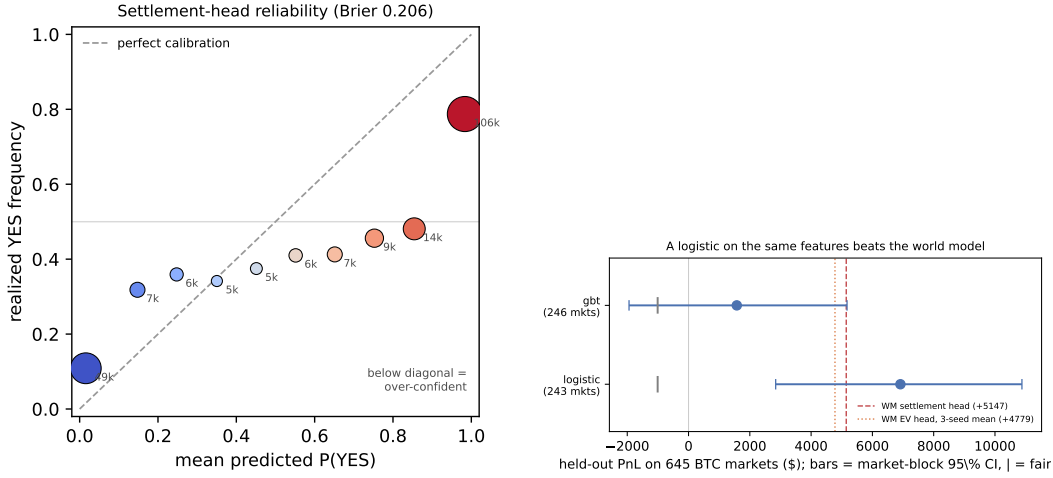


Figure 3: **Left:** settlement-head reliability over 214k held-out probabilities (Brier score 0.206, the mean-squared error between forecast probability and outcome; lower is better). Points fall below the diagonal for mean  $> 0.5$  (systematic over-confidence), yet the populous extreme deciles land on the correct side of 0.5 (0.787 top decile, 0.109 bottom decile; 0.785/0.108 under 0.1-wide bins). This is the directional signal the divergence trade harvests; mid-confidence bins (0.5–0.9) realize below 0.5, the calibration footprint of the low-predictability failure the EV head removes. **Right:** the economic edge does not need the architecture. Held-out PnL of the same-feature simple models on the 645-market set, with market-block 95% CIs. The logistic’s +\$6,914 exceeds both world-model heads, and its market-block interval (+\$2,844 to +\$10,880 over 243 independent markets) survives the dependence-respecting bootstrap. The market counts are model-specific traded/evaluable subsets under the same frozen gate: 243 for logistic and 246 for GBT, all drawn from the 645-market held-out BTC set.

385 **Economic-coda caveats (detail).** The edge harvests the lag between the underlying spot  
386 price and the contract book. Our backtest assumes each order executes at the prices recorded  
387 in the book, and that the resting liquidity there is available to trade against; in live trading,  
388 large orders would move the price against us and erode some of the edge. Only the  
389 train-frozen predictability gate is *confirmatory*; the depth-band and early-market cuts are  
390 *exploratory*, subject to forking-paths caveats. The reliability table and PnL totals are recorded-  
391 vintage structural diagnostics. The over-confidence pattern is structural, not vintage-specific.

392 **Regime robustness via a tradeable-EV head (detail).** We also supervise the *tradeable*  
393 expected value directly, with a `BookRelativeEdgeHead` trained under Huber loss on  $e_{yes} =$   
394  $outcome - ask_{yes}$  and  $e_{no} = (1 - outcome) - ask_{no}$ , where  $ask_{yes}$  and  $ask_{no}$  are the prices to

395 buy the YES and NO contracts, so the target is near zero (abstain) when the book is already  
 396 fair. This makes the low-predictability regime robust: low-pred +\$2,368 at  $P(\text{profit}) = 0.99$   
 397 (seed 0), and a three-seed mean of +\$4,779 total at mean low-pred  $P(\text{profit}) = 0.98$ , collapsing  
 398 the high-vs-low PnL spread  $6.2 \times$  (\$5,044  $\rightarrow$  \$814) at total PnL comparable to settlement  
 399 (+\$5,147). The same-feature logistic is *already* regime-robust (high-pred +\$2,448, low-pred  
 400 +\$4,466, both  $P(\text{profit}) \geq 0.99$ ), so the low-predictability signal is not intrinsically hard. The  
 401 EV head is a real fix *for the world-model readout*, not the only way to capture it.

## 402 **B Linear-probe localization of regime decodability**

403 Two probes localize the failure behind the null. A post-hoc reading of the jointly-supervised  
 404 router shows it is uniform *per step* (per-sample entropy  $\approx \log 4$ ) with near-chance agreement  
 405 against the efficiency-ratio bucket (0.25–0.31 vs. 0.25). Direct supervision does not even  
 406 make the router regime-discriminative. A frozen linear probe on the unmodulated features  
 407 decodes the per-step predictability bucket only weakly (0.33–0.36 vs. 0.25 chance) and the  
 408 volatility bucket somewhat better (0.43–0.47). Yet the volatility-axis gap is *also* null. So  
 409 even on the axis the representation encodes best, and even when the regime is supplied  
 410 by supervision, the in-scan modulation collapses to identity. (The degenerate next-tick  
 411 direction head is de-confounded by inverse-frequency class balancing, which yields a non-  
 412 degenerate macro-F1 with the gap still null; the single borderline gap, Kaggle spot-vol at  
 413  $n=2$ , is disproved by a pre-registered third seed that flips its sign.)